Incident Report: Repeat Seal Failure on Pump P-204

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Summary

On 2024-11-03 the mechanical seal on Pump P-204 failed for the second time in nine months. Leakage reached approximately 40 drops per minute before shutdown.

Operating conditions at failure

Parameter	Value
Process temperature (average)	105 deg C
Peak temperature recorded	112 deg C
Discharge pressure	7.5 bar
Seal face material	Carbon vs Silicon Carbide

Findings

Inspection found thermal cracking on the carbon seal face. The operating temperature exceeded the 90 deg C limit given in MAN-HX200-001 Revision B. No bearing damage was observed.

Root cause

Probable root cause: sustained operation above the rated temperature of the seal face material after the January 2024 process change. Contributing cause: the temperature change was not reviewed against the pump manual limits.

Corrective actions

1. Evaluate upgraded seal faces (action owner: Reliability). 2. Add temperature alarm at 95 deg C. 3. Review all assets affected by the process temperature change.